AI ELEVATE · SKY MILES LIMITED

BACSWN SkyWatch Bahamas

Live Data Demonstration Plan

What we will implement once the live weather data feed is provided — and what the demonstration will show.

Prepared for	Ministry of Transport & Aviation
Stakeholders	BCAA · NEMA · Department of Meteorology
Prepared by	AI Elevate (Sky Miles Limited — AI Elevate Division)
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COMMERCIAL IN CONFIDENCE

1 Purpose of This Demonstration

The BACSWN prototype is already live and feature-complete on simulated data. This demonstration is the step that turns “an impressive prototype” into **“a system we can trust with aviation safety and emergency response.”**

The single objective is to prove one thing to the people who will sign off on production:

> **Everything you see is driven by real, live data — and you can verify it yourself, in the room.**

> Simulated data demonstrates *features*. Real data demonstrates *trustworthiness* — and for a Meteorological Watch Office and an emergency-management agency, trust is the entire decision.

2 What We Need From You — Data Handover

To run the live demonstration we ask for access to the operational feeds BACSWN will consume in production. Each one is **already integrated** in the platform and currently running on a fallback.

Feed	Source / Owner	What it unlocks in the demo
METAR / TAF / SIGMET / PIREP	Dept. of Meteorology / AWC (Nassau FIR)	Live, verifiable aviation weather for MYNN, MYGF, MYEG and the 15-station network
Active weather alerts	NWS / regional alerting	Live hazard ingestion and corroboration
Forecast feed	Open-Meteo / Tomorrow.io	Forecast accuracy scoring against real outcomes
Tropical cyclone data	NHC (National Hurricane Center)	Live storm tracking and impact modeling
Flight surveillance	OpenSky / ADS-B (Nassau FIR)	Real airspace picture and live CORSIA emissions

> If any government feed cannot be released before the demo, we run that module on its current validated fallback and label it clearly as **SIMULATED** on screen. No module silently mixes real and simulated data.

3 What We Will Demonstrate

The agenda is organized by the three constituencies who must each say “yes.” Each item maps directly to a problem already identified in the BACSWN business case — the Dorian communications blackout, alert latency, the false-positive rate, and CORSIA compliance.

A · FOR THE METEOROLOGICAL WATCH OFFICE — THE “CAN I TRUST THE DATA?” TEST

- **Verifiable live METARs / TAFs.** The decoded observation for Nassau (MYNN) and other stations is shown beside the raw official product — a met officer can pull up the same product independently and confirm it matches, live.
- **Forecast verification scoreboard.** Because we hold both real observations and real forecasts, we show how accurate *yesterday’s* forecast actually was (bias, error, hit rate). This is the language a met office speaks — and almost no vendor demo shows it.
- **AI-drafted SIGMET against a real hazard.** When real convection or turbulence is present in the FIR, the system drafts an ICAO-format SIGMET from the live feed, shown next to what the office would issue — AI *assisting* the MWO’s statutory job, not replacing judgment.

B · FOR NEMA / EMERGENCY MANAGEMENT — THE “WILL IT HOLD WHEN DORIAN COMES BACK?” TEST

- **Live hurricane impact modeling.** Using the real NHC feed, we overlay the official forecast cone with our storm-surge flood zones, real shelter capacity, and evacuation routes for the actually-threatened islands. We can replay this against Hurricane Dorian (2019) to show what BACSWN would have advised for Abaco and Grand Bahama.
- **Graceful degradation, demonstrated live.** We show real data flowing, then simulate an upstream outage — and the system keeps the airspace and weather picture alive with a clear **DEGRADED** banner. Surviving a Category 5 communications blackout is the core promise of the proposed mesh network; here we prove the resilience behavior on real data.

C · FOR BCAA / COMPLIANCE & COMMERCIAL — THE “DOES IT PRODUCE SOMETHING FILEABLE?” TEST

- **Real CORSIA emissions report.** We take real flight tracks through the Nassau FIR, compute CO_2 using the ICAO CORSIA methodology, and produce a compliance-grade report — a regulatory artifact, not just a chart.
- **Multi-source data-quality check.** We show the same parameter (e.g. Nassau wind) from multiple independent sources agreeing — and flag disagreement automatically. This is the data-quality assurance behind the proposal’s “<2% false-positive” claim.

4 The Live Demonstration — Demo-Day Flow

We walk the room through one continuous story, moving from **trust** to **capability** to **operational confidence**:

1. **"Verify this yourself."** A real METAR you can check on your own phone. → ***I can trust the data.***
2. **"Here's how accurate we were yesterday."** Forecast verification scoreboard. → ***They understand operational meteorology.***
3. **"Watch the AI draft the exact advisory your office would issue."** Live SIGMET. → ***This does our hard work.***
4. **"Here's the live storm impact on the islands at risk."** Hurricane + surge + shelters + evacuation. → ***This is built for our worst day.***
5. **"And here's the CORSIA report you could file tomorrow."** Compliance artifact. → ***This is operational and revenue-relevant.***
6. **"Now watch a data source fail."** Degraded-mode resilience. → ***It stays up when it matters most.***

Total run time: approximately 30–40 minutes plus Q&A.

5 Timeline — Data Handover to Demonstration

Phase	Duration	Activity
Phase 0 Data handover	Day 0	Credentials / feed access provided for the feeds in Section 2
Phase 1 Integration & validation	Days 1–3	Flip each feed from fallback to live; cross-check parsed output against official products
Phase 2 Demo hardening	Days 4–5	Production run mode, verification scoreboard, on-screen real / simulated labeling, rehearsal
Phase 3 Live demonstration	Demo day	On-site or remote briefing to Ministry / BCAA / NEMA

> Most integrations are flip-a-switch — the real-data paths already exist and are tested. The build effort is concentrated in the verification scoreboard and presentation polish.

6 What "Success" Looks Like

The demonstration has succeeded if, by the end, each stakeholder can answer "yes":

Stakeholder	The “yes” we are earning
Met Office	“I independently verified the data, and the SIGMET drafting matches our practice.”
NEMA	“I saw it model a real storm’s impact on our islands, and saw it survive a data outage.”
BCAA / Commercial	“It produced a compliance artifact we could actually use.”
Ministry	“This is a working operational system on live data — not a slideshow.”

7 Why This De-Risks the Production Decision

This demonstration is the lowest-cost, highest-confidence checkpoint before committing to the phased production program already proposed. It proves **on real Bahamian data** the exact capabilities the production network will deliver at scale — verifiable accuracy, ICAO-compliant advisory generation, hurricane resilience, and CORSIA compliance — before any field hardware is deployed.

It converts the proposal’s promises into something the decision-makers have watched work with their own data.